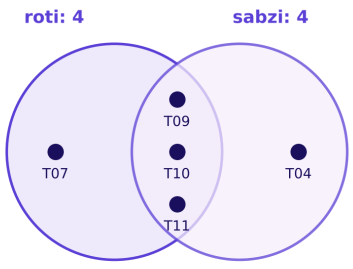
CHEAT SHEET

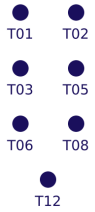
Which pairs beat chance, and what is each one worth?

Week 4 Wednesday. The five topics on a hostel canteen's twelve lunch orders and a Pune corner shop's month, both invented, then on Swiggy Instamart's Pune carts, whose strip fell from 6.1 to 3.8 percent click-through once it showed the items most often found in the same cart. Illustrative case: the company is real, the situation and the data are synthetic, built for this class.

C2_W04_D03_beyond_chance_STUDENT
Verified 13 October 2026
7 panels, 1 diagram

1 Support, confidence and lift, derived by counting: what does each answer?





twelve lunch orders, invented

	sabzi	no sabzi	all
roti	3	1	4
no roti	1	7	8
all	4	8	12

support 0.25 **confidence 0.75** **lift 2.25**

Count	It answers	Formula	Roti and sabzi
Support	How often together?	both / all	3 / 12 = 0.25
Confidence	How reliably, from A?	both / A	3 / 4 = 0.75
Lift	How far above chance?	both x all / (A x B)	3 x 12 / (4 x 4) = 2.25

Crux. Lift is the baskets seen holding both over the baskets chance gives, the same from either side; it counts a pull and never names a cause.

2 The confidence trap; lift near one: who is the worst partner?

Toward chaas, in 0.75 of orders	Confidence	Lift
dal rice, sabzi	1.00	1.33
roti	0.75	1.00
papad	0.60	0.80

Milk is in 71 percent of Instamart's carts; lift with milk runs 0.99 to 1.04.

Crux. The item in every basket is the worst bundle partner, because it was going in anyway.

3 Basket analysis on the order items: how do lines make pairs?

```
baskets = lines.groupby("order_id").item.apply(set)
pairs = Counter(p for b in baskets
                for p in combinations(sorted(b), 2))
```

k items make k x (k - 1) / 2 pairs: 33 lines make 30; unsorted keys count 60.

Crux. Sort the names so each pair has one key; the table holds counts, never a ranking.

4 The support floor and the watch list: when is a pair too thin?

Region	Rule	Canteen	Instamart
Promote	floor met, lift 1.5 up	roti and sabzi, 2.25 on 3	chips and cold drink, 3.39
Watch	lift 1.5 up, below floor	gulab jamun and curd, 6.0 on 1	ice cream and sauce, 4.60 on 100
Trap	floor met, near one	dal rice and chaas, 1.33 on 6	milk's pairs, 0.99 to 1.04

Crux. A floor is a count set first, 3 orders or 400 carts; pure chance reads 8.57 on 2 orders.

5 Cross-sell and upsell reasoning; expected value in rupees: what is a combo worth?

```
value = margin on purchases it changes - discount on every bill
- fixed cost
```

Sharma Kirana: 18 packs x Rs 12 = Rs 216, less Rs 168 handed back, less Rs 20 = Rs 28 a month; at 2 percent, -Rs 71. Rusk sales on every combo bill read Rs 5,040.

Crux. Cross-sell adds a second item, upsell a dearer one; each earns only on what it changes.

6 The case: what did Instamart's carts say?

Bread to milk read confidence 1.00 and lift 1.50 on twelve carts, and 0.74 and 1.04 on 40,000. The fall: 0.5 x 1.2 + 0.5 x 6.4 = 3.8 percent. A slot earns Rs 17,280 a day against Rs 3,240 showing milk; re-ranking recovers about Rs 6.3 lakh a month.

Crux. Say the base rate, the floor and the cost line, and end on one decision.

7 How do the five topics join?

Three counts (0.25, 0.75, 2.25), then the trap (1.33 behind a confidence of 1.00), then the pairs table (30 pairs from 33 lines), then the floor (3 orders), then the rupees (Rs 28): each counted on the canteen or the kirana first.

Crux. Every topic was counted by hand before Instamart's carts used it.

THE WORDS ON THIS SHEET

Support Share of all baskets holding both items

Confidence Share of A's baskets that also hold B

Lift Baskets with both, over the baskets chance alone gives

Base rate How common an item is across all baskets

Support floor The fewest baskets a pair needs before its lift is read

Watch list A strong, thin pair held for a trial with a pass rule